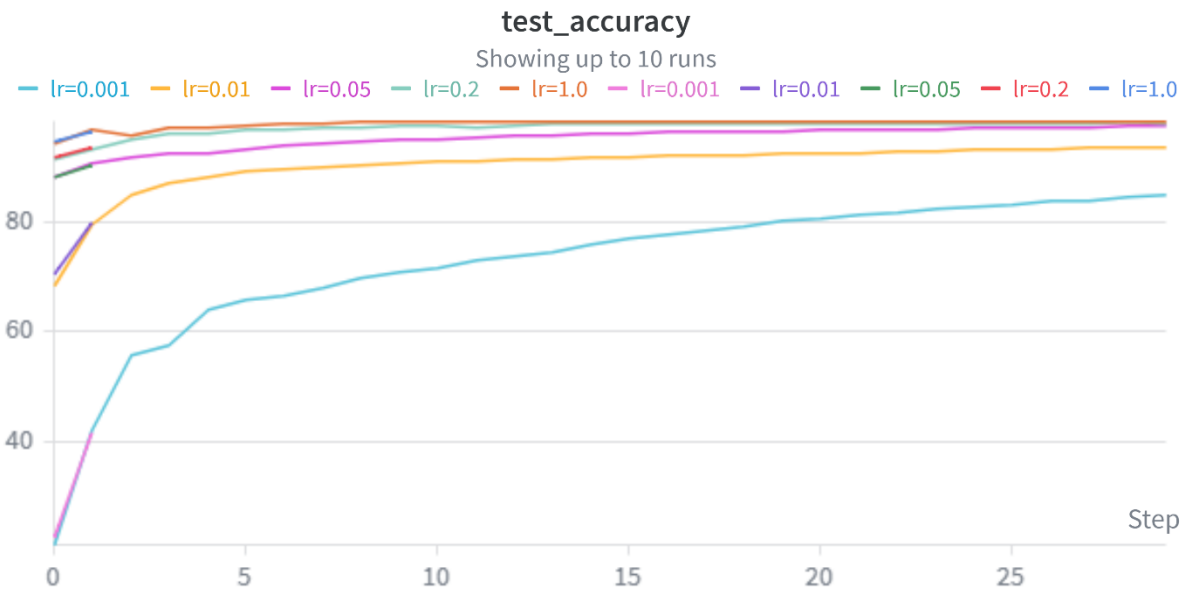
Exercise 3

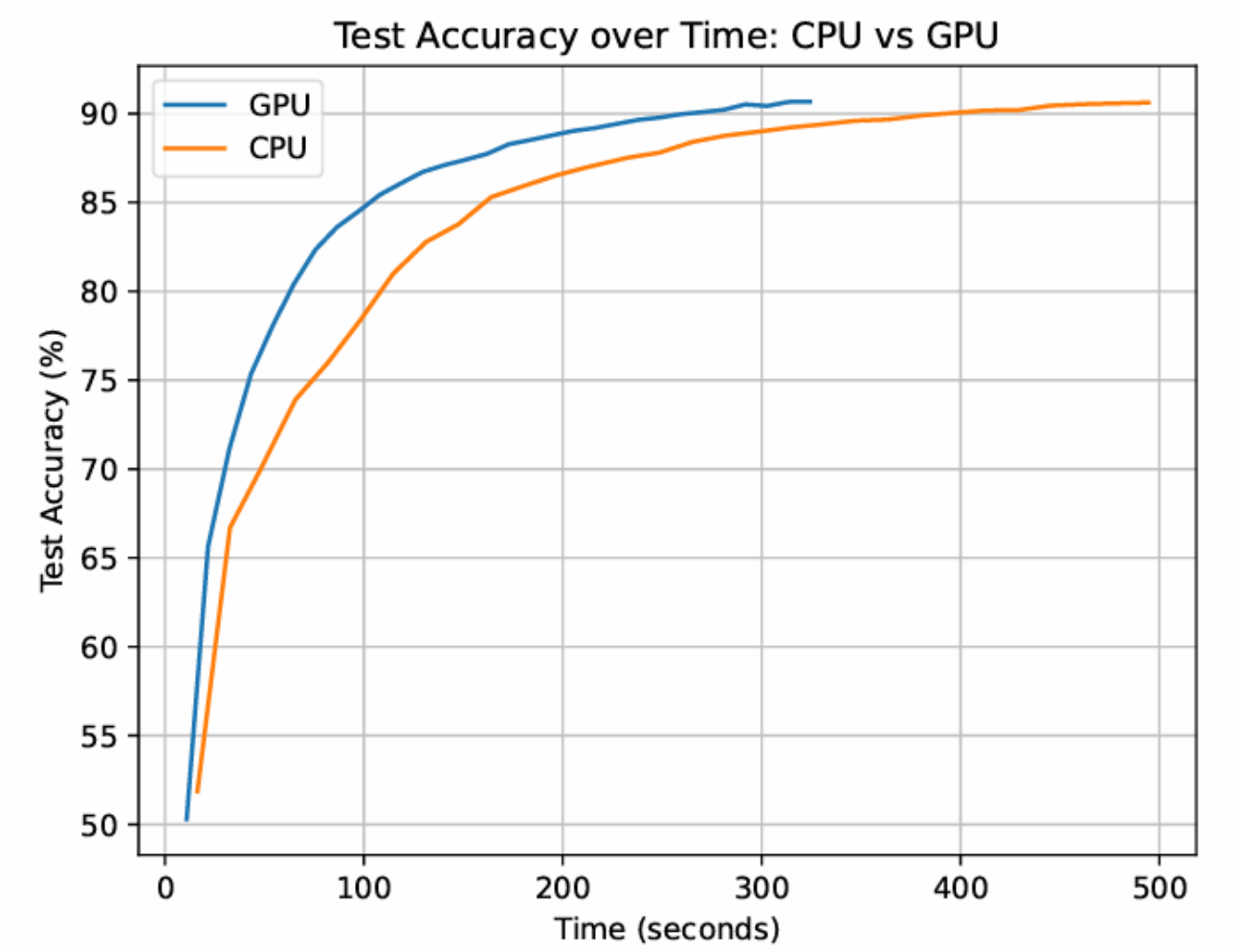
Luka Zotovikj

Aleksandar Toslev

3.1 Experiment logging


test_accuracy
Showing up to 10 runs
lr=0.001
lr=0.01
lr=0.05
lr=0.2
lr=1.0
lr=0.001
lr=0.01
lr=0.05
lr=0.2
lr=1.0
80
60
40
Step
0
5
10
15
20
25

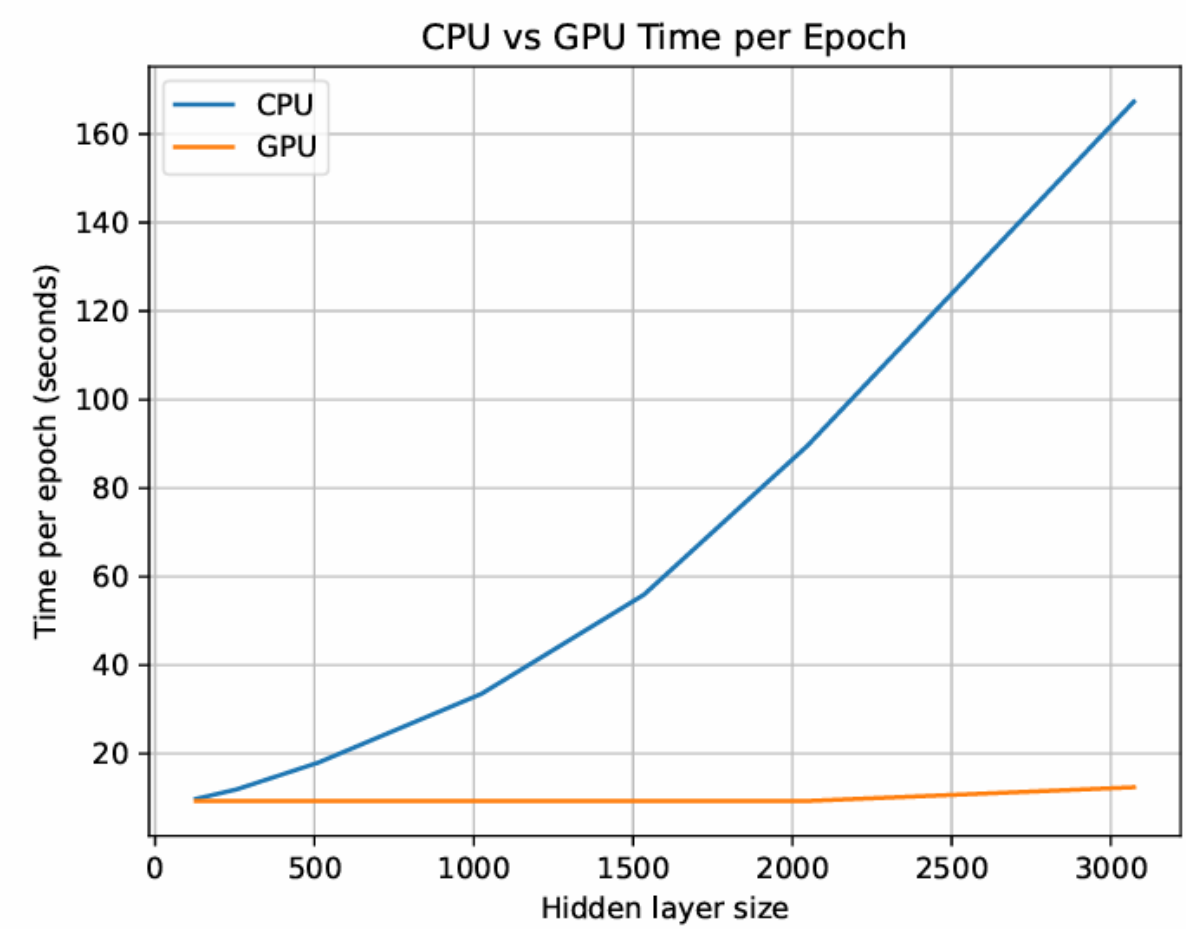
3.2 Neural network on GPU



In this plot we compare the test accuracy of the CPU and GPU implementation over time.

We see that both implementations reach a test accuracy of ~90%, so the model behaves similarly on CPU and GPU. The only difference is mainly the time needed to reach the 90%. The GPU curve rises faster and reaches high accuracy earlier, while the CPU needs more time for the same progress. This makes sense because the neural network uses many matrix operations, which can be computed more efficiently in parallel on the GPU.

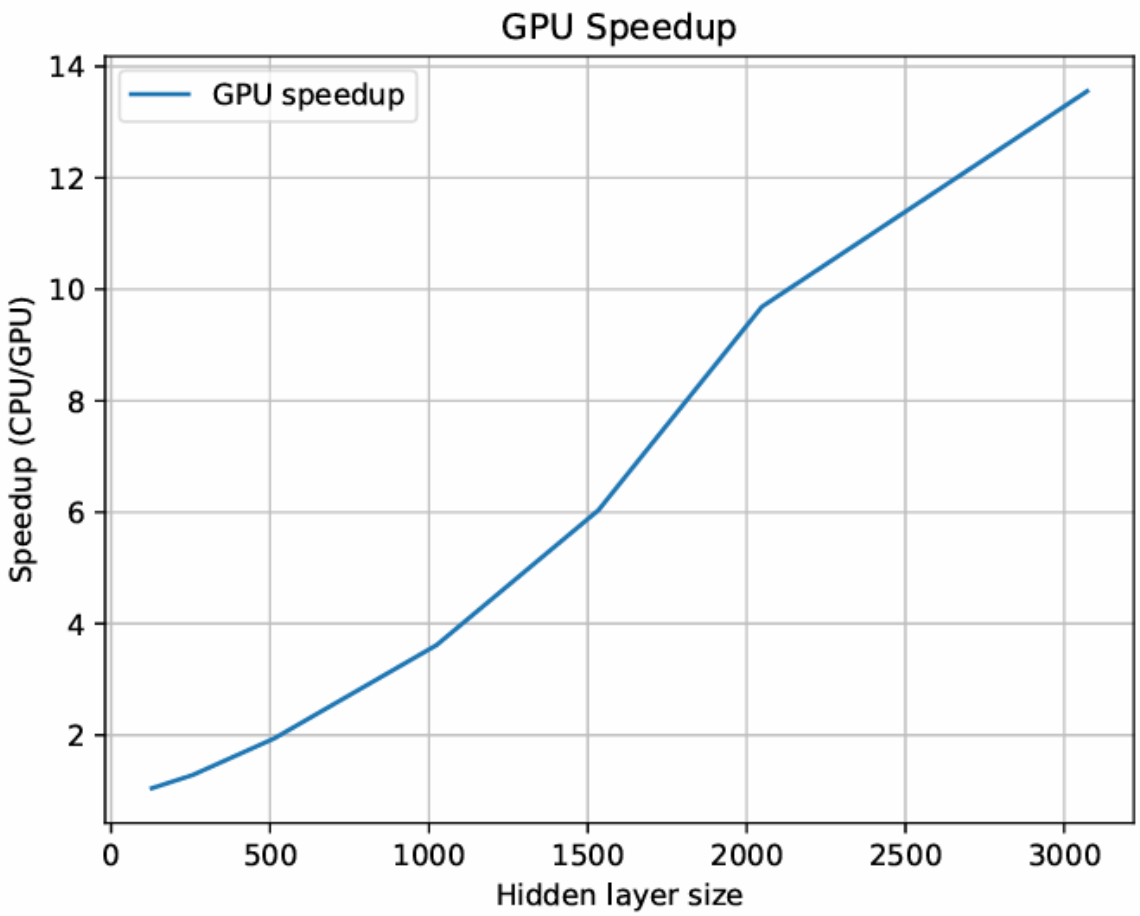
To conclude, the difference is visible, but not extremely huge, because there is also some overhead from data transferring.



In this plot we can see that the CPU runtime increases a lot when the layer size becomes larger. This is expected, because bigger layers mean bigger weight matrices and therefore more expensive matrix multiplications.

For observe that the GPU time stays almost constant for the smaller and medium layer sizes and there is a little increase for the largest layer size.

We conclude that the GPU handles the larger matrix operations much better than the CPU.



In the speedup plot we observe that the GPU has a bigger advantage when the network (layer) size increases. For small hidden layers the speedup is close to 1, so the GPU is only slightly faster than the CPU. This is likely because the overhead of moving data and starting GPU computations takes some amount of time.

For larger layers the speedup grows strongly, up to around 13–14 times faster than the CPU at around 2700-3000 layer size.

We conclude that the GPU becomes really useful when the model and matrix operations are large enough.